



秀傳醫療財團法人彰濱秀傳紀念醫院
Chang Bing Show Chwan Memorial Hospital

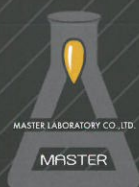
檢測報告

Test Report



Chang Bing Show Chwan





瑞德生物科技有限公司 MASTER LABORATORY CO.,LTD.

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瑞德生物科技有限公司
MASTER LABORATORY CO., LTD.

Acute Systemic Toxicity Study

Master Laboratory Co., Ltd. Animal Laboratory

Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township,

Hsinchu County 31053, Taiwan (R.O.C.) TEL: (886-2) 25176117



**RELANO WOUND SPRAY
Acute Systemic Toxicity Study
STUDY REPORT**

Sponsor : Nanoplus Life Biomedical Technology Co., Ltd.

Testing Institution : Master Laboratory Co., Ltd.

July 2023



FINAL REPORT

Report No.: 23S186T04-01-R01

Report No.: 23S186T04-01-R01

Experimental starting date: 06.15.2023

Test article registration date: 06.16.2023

Animal in-housing: 06.15.2023

Extraction of test article: 06.23.2023

Experimental administration date: 06.26.2023

Observation of toxicity reaction: 06.26.2023-06.29.2023

Animal sacrifice date: 06.29.2023

Experimental complete date: 06.29.2023

Study Announcement

1. The study report is valid for the test article used only, and shall not be partly recopied or extracted for another object.
2. The study report is invalid without the endorsement of Master Laboratory Co., Ltd.



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SIGNATURE OF STUDY PERSONNEL

Study Director

Xin Han Huang
Xin Han Huang

07, 17, 2023
Date

Facility Management

Alan Hsieh
Alan Hsieh

07.17.2023
Date



GLP COMPLIANCE STATEMENT

The study met with the technical requirements of the protocol, and the applicable guidance, which included the Good Laboratory Practice for Non-Clinical Laboratory Studies (FDA, 21 CFR, Part 58, 2019), OECD Principles of Good Laboratory Practice (No. 1, 1997) and Good Laboratory Practice for Non-Clinical Laboratory Studies (Food and Drug Administration, R.O.C., 2019), except the parts of test article identification and related analyses (21 CFR §58.105, §58.113, FDA; 6.2.2-6.2.5, OECD GLP No.1, 6.2.2-6.2.5, TFDA GLP). To the best of our knowledge, there were no deviations from the approved study plan and no adverse problem that would affect the integrity of this study or the interpretation of the study result.

Study Director

Xin Han Huang
Xin Han Huang

07, 17, 2023
Date



QUALITY ASSURANCE STATEMENT

The Quality Assurance personnel had inspected the conduct of different phases of the study according to a predetermined study schedule. To the best of our knowledge, there were no deviations from the study plan and standard operating procedures that would affect the integrity of this study. This report has been audited by the QA personnel in accordance with the appropriate standard operating procedures of Master Laboratory Co., Ltd. described the methods and procedures used in the study, and the reported results accurately reflect the raw data generated during this study.

Listed below are the phases in this study that were audited by the QA personnel and the dates the audits were performed and findings reported to facility management (FM).

Inspection record:

Date of inspection	Phase	Date Reported to SD	Date Reported to FM
06.12.2023	Study protocol, Personnel quality and test article. (Study base)	06.12.2023	06.12.2023
07.03.2023	Study system, study report, raw data, and final report. (Study base)	07.03.2023	07.03.2023

Quality Assurance unit in charge

Ying Chun Chen
Ying Chun Chen

07.17.2023
Date



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Contract Research Organization and Sponsor Information

1. CRO:

- a. Title: Master Laboratory Co., Ltd.-Animal laboratory
- b. Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan, R.O.C.

2. Study Director:

- a. Name: Xin Han Huang
- b. Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan, R.O.C.

3. Study personnel:

- a. Name: Xin Han Huang
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4. Quality Assurance personnel:

- a. Name: Ying Chun Chen
- b. Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan, R.O.C.

5. Sponsor:

- a. Title: Nanoplus Life Biomedical Technology Co., Ltd.
- b. Address: 9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)



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SUMMARY

The present study was to investigate the biological reactivity of the test article “RELANO WOUND SPRAY” extracts in mice. The testing was performed in compliance with ISO 10993-11:2017 guidance for systemic toxicity study. In control groups, animals were injected intravenously with saline and injected intraperitoneally with cottonseed oil in the dosage 50 ml/kg. In treatment groups, animals were injected with polar test article extracts intravenously and with non-polar test article extracts intraperitoneally in the same dosage. All of test and control animals were in overall good health over the course of the study, and none of the animals treated with the test article extract showed significantly greater reaction than the control animals. All animals survived to their scheduled time point, the final body weights of animals in treatment group were no significant difference compared with the control group. The results showed that there were no significant clinical signs and gross findings in either the control or treatment groups (Table1 and Appendix 1). Therefore, the test article extracts did not cause toxicity reaction or death after intravenous and intraperitoneal injection in this acute systemic toxicity study.



INTRODUCTION

The study was performed in compliance with ISO10993-11:2017 guidance for systemic toxicity study. The present study was to evaluate the biological reactivity after injecting the test article extracts into the mice.

MATERIALS AND METHODS

1. Animals

1.1. Species/Strain: ICR mice

1.2. Resource: BioLASCO Taiwan Co., Ltd

1.3. Body weights (Gender):

According to ISO10993-11:2017, the weight variation of animals used should not exceed $\pm 20\%$ of the mean weight (Male).

1.4. Quarantine/acclimation :

Animals were subjected to be quarantined and acclimated before treatment.

Veterinarian ensured the animal health status before the treatment.

1.5. Reasons chosen for animal experimentation:

According to ISO10993-11:2017 guidance, ICR mice were proven to be suitable for acute systemic toxicity test.

1.6. Groups

Groups	Control	Treatment
Number of animals	5	5
Treated article	Saline	RELANO WOUND SPRAY Saline extracts
Number of animals	5	5
Treated article	Cottonseed oil	RELANO WOUND SPRAY Cottonseed oil extracts



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2. Feeding and care

2.1. Housing: Mice room

2.2. Environment

a. Temperature: $22 \pm 3^{\circ}\text{C}$

b. Humidity: $50 \pm 20\%$

c. Light Cycle: 12 hours light and 12 hours dark

2.3. Cage and animal number

a. Quarantine/acclimation: 5 mice/cage

b. Study period: 5 mice/cage

2.4. Feed

a. Name: Lab Diet[®] 5001 Rodent Diet

b. Brand: Lab Diet

c. Way to supply: *ad libitum*

d. Source: PMI Nutrition International, U.S.A.

2.5. Drinking water

a. Sort: RO Water

b. Way to supply: *ad libitum*

3. Individual and group identification

3.1. Individual identification: Tested animals were identified by ear-marking.

3.2. Group identification: Cages were properly labeled for identification including the Study Title/No., Administration, Observation Period, Room No., Cage No., Quantity/cage, Species, Strain, Gender, In House Date, In House Age, Animal ID No., Keeper and Deputy.



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4. Test article and control vehicles

4.1. Test article: RELANO WOUND SPRAY, Lot. 51223050003

4.2. Polar control vehicle (Saline): CHI SHENG CHEMICAL CORPORATION, Lot. QB176; Solutio Natrii Chloridi Isotonica 0.9% 500 ml

4.3. Non-Polar control vehicle (Cottonseed oil): SIGMA-ALDRICH, Lot. MKCR3879, CAS No. 8001-29-4

5. Administration of test article and control vehicles

5.1. Preparation : According to ISO 10993-12:2021 guideline.

a. Polar preparation: Measured the weight of test article, and immersed it in saline for 72 ± 2 hr at $50 \pm 2^\circ\text{C}$ with constant agitation (100 rpm) with an extraction ratio of 0.2 g/ml. After extraction, the extract was used immediately; the appearance of test article extract was clear, particulate-free and no color changed.

b. Non-polar preparation: Measured the weight of test article, and immersed it in cottonseed oil for 72 ± 2 hr at $50 \pm 2^\circ\text{C}$ with constant agitation (100 rpm) with an extraction ratio of 0.2 g/ml. After extraction, the extract was used immediately; the appearance of test article extract was clear, particulate-free and no color changed.

5.2. Method, route and frequency of administration :

Single dose of test article extract and control vehicles were injected into each mouse. Polar extraction vehicle was injected intravenously and non-polar extraction vehicle was injected intraperitoneally, respectively.

5.3. Dosage of administration: Dosage of test article extracts or control vehicles were 50 ml/kg.

5.4. I.V. injection administration dose rate: 1.0 ml/min.



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6. Procedure

- 6.1. Polar groups were injected by I.V. injection and non-polar groups were injected by I.P. injection.
- 6.2. All mice were observed immediately after injection and again at 4th, 24th, 48th and 72nd hour after injection.

7. Animal observations and items for examination

7.1. Determination of toxicity reaction:

- 7.1.1. Post-injection, the body weight at 24th, 48th and 72nd hour were recorded.
- 7.1.2. The changes in skin, fur, somatomotor activity, behavior pattern, and the clinical observation according to “Common clinical signs and observations” (Appendix 4) were recorded by cage-side observations at 4th, 24th, 48th and 72nd hour.

7.2. Evaluation criteria: According to ISO 10993-11: 2017 guideline.

- 7.2.1. If during the observation period of an acute systemic toxicity test none of the animals treated with the test sample shows a significantly greater biological reactivity than animals treated with the vehicle control, the sample meets the requirements of this test.
- 7.2.2. Using five animals, if two or more animals die, or if behaviour such as convulsions or prostration occurs in two or more animals, or if a final (end of study) body weight loss greater than 10 % occurs in three or more animals, the sample does not meet the requirements of the test. Any transitory body weight loss should be critically evaluated along with other clinical observations in the assessment of systemic toxicity.



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8. Archive of the record and data

8.1. All records and data were preserved for six years. Test articles (retained samples) were retained for six years excluding the samples that sponsors require to be fully returned. Archive address: 3F., No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan (R.O.C.).



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RESULTS

The study was to investigate the biological reactivity of the test article “RELANO WOUND SPRAY” extracts in the mice. In control groups, animals were injected intravenously with polar extraction vehicle saline and injected intraperitoneally with cottonseed oil in the dosage 50 ml/kg. In treatment groups, animals were injected with polar test article extract intravenously and non-polar extract intraperitoneally with the same dosage. After injection, the systemic responses at 4th, 24th, 48th and 72nd hour were recorded and evaluated. Test article group showed no more significant biological reactivity than control group (Table 1 and Table 2) and no body weight loss in all treatment animals (Appendix 3), which indicate that the test article met the requirements of the acute systemic toxicity study.

The results showed that there were no significant clinical signs and gross findings in either the control or treatment groups. Therefore, the test article extracts did not cause toxicity reaction or death in the mice.

CONCLUSION

The present study was performed in compliance with ISO 10993-11:2017 guideline for systemic toxicity study. All of test and control animals were in overall good health over the course of the study, and none of the animals treated with the test article extract showed significantly greater reaction than the control animals. All animals survived to their scheduled time point, the final body weights of animals in treatment group were no significant difference compared with the control group. In addition, final body weight loss in the animals of the treatment and control groups were not greater than 10% of the original body weights, which all met the criteria showed in ISO10993-11:2017. Therefore, the test article “RELANO WOUND SPRAY” extracts did not cause toxicity reaction in this acute systemic toxicity study.



REFERENCES

1. Good Laboratory Practice for Nonclinical Laboratory Studies (2019) Food and Drug Administration, R.O.C.
2. Good Laboratory Practice for Nonclinical Laboratory Studies. Title 21 of the U.S. Code of Federal Regulations, Part 58 (2019) United States Food and Drug Administration.
3. Biological evaluation of medical devices-Part 11: Tests for systemic toxicity ISO 10993 (2017).
4. Biological evaluation of medical devices-Part 11: Tests for systemic toxicity EN ISO 10993 (2018).
5. Biological evaluation of medical devices-Part 12: Sample preparation and reference materials ISO 10993 (2021).
6. Biological evaluation of medical devices-Part 12: Sample preparation and reference materials EN ISO 10993 (2021).
7. General requirements for the competence of testing and calibration laboratories. ISO/IEC 17025 (2017).
8. Standard Practice for Evaluating Material Extracts by Systemic Injection in the Mouse. ASTM F750-87 (2012).



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Table 1. Incidence of Clinical Observation in mice

Group (Saline)	Control	Treatment
Number of animals	5	5
Treated article	Saline	RELANO WOUND SPRAY Saline extracts
Toxicity reaction	0/5	0/5
Death	0/5	0/5
n/n : No. of mice with clinical signs/No. of mice per group		
Group (Cottonseed oil)	Control	Treatment
Number of animals	5	5
Treated article	Cottonseed oil	RELANO WOUND SPRAY Cottonseed oil extracts
Toxicity reaction	0/5	0/5
Death	0/5	0/5
n/n : No. of mice with clinical signs/No. of mice per group		



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Table 2. Incidence of gross finding

Group (Saline)	Control	Treatment
Dosage (ml/kg)	50	50
Gender	Male	Male
Animal number	5	5
Symptom	0/5	0/5
n/n : No. of mice with gross signs/No. of mice per group		
Group (Cottonseed oil)	Control	Treatment
Dosage (ml/kg)	50	50
Gender	Male	Male
Animal number	5	5
Symptom	0/5	0/5
n/n : No. of mice with gross signs/No. of mice per group		



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Appendix 1-1. Individual Clinical Observation of Mice

Injection route (intravenous)	Treated article	Gender	Animal ID	Items for Grading	Clinical Observation (time point/h)			
					4	24	48	72
Control group	"Saline"	Male	1001	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1002	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1003	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1004	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1005	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
Treatment group	RELANO WOUND SPRAY "Saline" extracts	Male	1006	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1007	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1008	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1009	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1010	Toxicity reaction	-	-	-	-
				Death	-	-	-	-

One clinical observation



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Appendix 1-2. Individual Clinical Observation of Mice

Injection route (peritoneal)	Treated article	Gender	Animal ID	Items for Grading	Clinical Observation (time point/h)			
					4	24	48	72
Control group	"Cottonseed oil"	Male	1011	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1012	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1013	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1014	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1015	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
Treatment group	RELANO WOUND SPRAY "Cottonseed oil" extracts	Male	1016	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1017	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1018	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1019	Toxicity reaction	-	-	-	-
				Death	-	-	-	-
		Male	1020	Toxicity reaction	-	-	-	-
				Death	-	-	-	-

None clinical observation



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Appendix 2. Individual Animal in Clinical Observation

Polar Group	Dosage (ml/kg)	Gender	Animal ID	Descriptions of Clinical Symptoms
Control	50	Male	1001	None clinical symptoms
			1002	None clinical symptoms
			1003	None clinical symptoms
			1004	None clinical symptoms
			1005	None clinical symptoms
Treatment	50	Male	1006	None clinical symptoms
			1007	None clinical symptoms
			1008	None clinical symptoms
			1009	None clinical symptoms
			1010	None clinical symptoms
Non-Polar Group	Dosage (ml/kg)	Gender	Animal ID	Descriptions of Clinical Symptoms
Control	50	Male	1011	None clinical symptoms
			1012	None clinical symptoms
			1013	None clinical symptoms
			1014	None clinical symptoms
			1015	None clinical symptoms
Treatment	50	Male	1016	None clinical symptoms
			1017	None clinical symptoms
			1018	None clinical symptoms
			1019	None clinical symptoms
			1020	None clinical symptoms

None clinical symptoms: All of test and control animals were in overall good health over the course of the study, and none of the animals treated with the test article showed significantly greater reaction than the control animals.



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Appendix 3. Individual Body Weight

Polar Group	Dosage (ml/kg)	Gender	Animal No.	Body weight (g)				
				0hr	24hr	48hr	72hr	+/-
Control	50	Male	1001	30.1	31.1	31.8	32.6	+2.5
			1002	29.0	30.0	31.0	31.8	+2.8
			1003	29.0	30.1	30.9	31.4	+2.4
			1004	30.6	31.9	32.1	33.5	+2.9
			1005	28.8	30.0	30.5	31.5	+2.7
Treatment	50	Male	1006	26.6	27.8	28.2	28.6	+2.0
			1007	26.9	27.9	28.5	28.9	+2.0
			1008	30.1	30.9	31.7	32.4	+2.3
			1009	25.8	27.5	28.0	28.6	+2.8
			1010	28.7	30.3	30.9	31.7	+3.0
Non-Polar Group	Dosage (ml/kg)	Gender	Animal No.	Body weight (g)				
				0hr	24hr	48hr	72hr	+/-
Control	50	Male	1011	28.0	28.3	28.9	29.5	+1.5
			1012	30.6	30.7	31.0	31.2	+0.6
			1013	28.1	28.1	28.8	29.5	+1.4
			1014	27.0	27.1	28.6	29.2	+2.2
			1015	27.9	29.2	29.5	29.9	+2.0
Treatment	50	Male	1016	26.8	27.2	28.0	28.8	+2.0
			1017	27.3	28.0	28.8	29.6	+2.3
			1018	27.7	26.6	27.5	28.1	+0.4
			1019	26.1	25.6	26.3	26.5	+0.4
			1020	26.5	27.0	28.3	29.1	+2.6



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Appendix 4. Common clinical signs and observations (ISO10993-11)

Clinical observation	Observed sign	Involved system(s)
Respiratory	Dyspnea (abdominal breathing, gasping), apnoea, cyanosis, tachypnea, nostril discharges	CNS, pulmonary, cardiac
Motor activities	Decrease/increase somnolence, loss of righting, catalepsy, ataxia, unusual locomotion, prostration, tremors, fasciculation	CNS, somatomotor, sensory, neuromuscular, autonomic
Convulsion	Clonic, tonic, tonic-clonic, asphyxial, opisthotonos	CNS, neuromuscular, autonomic, respiratory
Reflexes	Corneal, righting, myotactic, light, startle reflex	CNS, sensory, autonomic, neuromuscular
Ocular signs	Lacrimation, miosis, mydriasis, exophthalmos, ptosis, opacity, iritis, conjunctivitis, chromodacryorrhea, relaxation of nictitating membrane	Autonomic, irritation
Cardiovascular signs	Bradycardia, tachycardia, arrhythmia, vasodilation, vasoconstriction	CNS, autonomic, cardiac, pulmonary
Salivation	Excessive	Autonomic
Piloerection	Rough hair	Autonomic
Analgesia	Decrease reaction	CNS, sensory
Muscle tone	Hypotonia, hypertonia	Autonomic
Gastrointestinal	Soft stool, diarrhoea, emesis, diuresis, rhinorrhoea	CNS, autonomic, sensory, GI motility, kidney
Skin	Oedema, erythema	Tissue damage, irritation



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Appendix 5. Test Article Information Sheet

Master Laboratory Co., Ltd.

Information for ☒ Test Article / ☐ Control Article

Sponsor Company	Nanoplus Life Biomedical Technology Co., Ltd.
Sponsor Address	9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)
Contract Study item	☒ Base on the contract
Name of test article	RELANO WOUND SPRAY (良諾液態保護噴劑)
Major components	Nano Ion Water, Sodium Carboxymethylcellulose
Sample status	☐ Sterilized (☐ Gamma ☐ EO ☐ Steam) ☒ Not Sterilized
Storage condition	☒ Room temperature (10°C-30°C) ☐ Refrigeration (2°C-8°C) ☐ Others:
Expiry day	05 / 26 / 2025
Specific requirement	None.
Lot number	☒ Base on the specific number on the package: LOT51223050003 ☐ Others:
Extract by	☒ Weight (0.2g/ml) Total weight of each test article: <u>about 50 g</u> (1g/ml) ☐ Surface (Sample thickness: ☐ >1.0mm ☐ 0.5-1.0mm ☐ <0.5mm) Total area surface of each test article:
Absorption	☒ Non absorption ☐ Water absorption rate:/ Oil absorption rate:
Sponsor Signature	



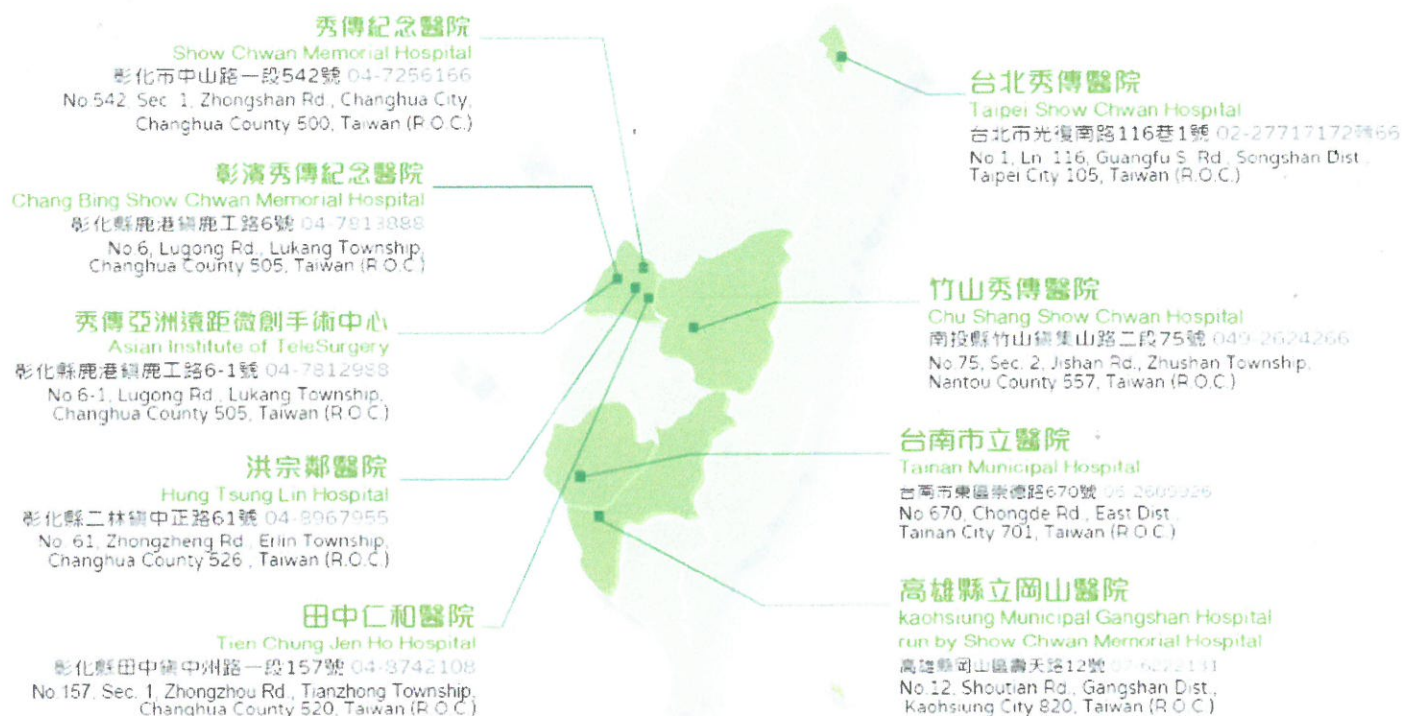
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Appendix 6. Test Article







秀傳醫療財團法人彰濱秀傳紀念醫院
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